

CSDS 302 Discrete Math | Fall 2026 | Homework 8

Instructions

Section 1.4: 5, 8, 14, 17

Section 10.1: 2ad, 6bc, 7, 8bc, 9bc, 13, 20, 22, 26, 29

Section 10.2: 2b, 3b, 4bd, 19bc, 22a

—

Feedback

Score: 144/152 (94.74%)

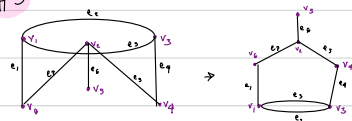
Section 10.1

8b: -4, should be 2

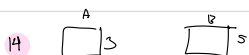
8c: -4, should be 3

1.4: 5, 8, 14, 17

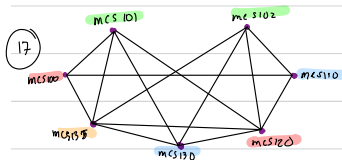
#5



- H8 (i) e_1, e_2, e_3 (v) e_8, e_9, e_{10}, e_{11}
 (ii) v_1, v_2, v_3 (vi) v_4, v_5
 (iii) e_1, e_2, e_3, e_4 (vii) S
 (iv) e_6, e_7

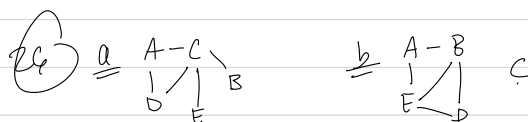
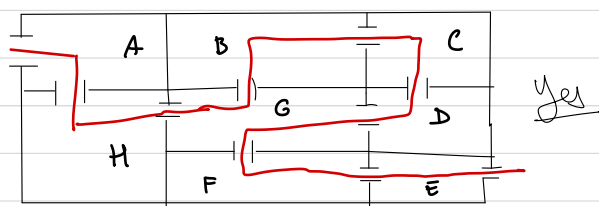


- 1) Fill up sq.
 2) Pour A to B so B has 3
 3) Fill A, Fill B to top.
 B has 5 & A has 1



(20) No Euler trail. E & H have odd degrees

(22)



(29) ALL VERTICES ONCE
 $V_0, V_7, V_1, V_2, V_3, V_5, V_6, V_8$

10.2: 2b, 3b, 4bd, 19bc, 22a

(2)

	V_1	V_2	V_3	V_4
V_1	1	0	1	0
V_2	0	0	1	0
V_3	1	0	0	1
V_4	0	0	1	0

(3b)

	V_1	V_2	V_3	V_4
V_1	0	1	0	0
V_2	0	0	1	0
V_3	1	2	1	0
V_4	0	0	1	0

10.1: 2ad, 6bc, 7, 8bc, 9bc, 13, 20, 22, 26, 29

(2) a. walk d. trail

(6) b. $V_1, V_2, V_3, V_4, V_7, V_8$

c. $V_2, V_3, V_6, V_7, V_8, V_9, V_{10}, V_{11}$

(7) a. N edges, $n+1$ vert

1 2 ... $n+1$ straight line graph
 any edge will disconnect.
 $\Rightarrow N$ edges n vertices
 NOTHING OCCURS

(8) $\cong 1 \cong 2$

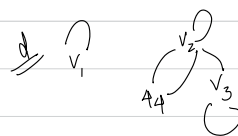
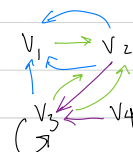
(9) $\cong$ Yes. All v have even degrees

$\cong$ No. G can be disconnected $\therefore$ No Euler circuit

degree	degree	degree	
V_1, V_3	V_4	V_7	V_1 has odd degree no Euler circuit
V_2	V_5	V_8	
V_3	V_6	V_9	

(4) b

	V_1	V_2	V_3	V_4
V_1	1	0	0	0
V_2	0	1	1	2
V_3	0	1	1	0
V_4	0	2	0	0



	V_1	V_2	V_3	V_4	V_5
V_1	0	0	0	1	2
V_2	0	0	0	1	1
V_3	0	0	0	2	1
V_4	1	1	2	0	0
V_5	2	1	1	0	0

$$A = \begin{bmatrix} V_1 & V_2 & V_3 \\ V_1 & 1 & 1 & 2 \\ V_2 & 1 & 0 & 1 \\ V_3 & 2 & 1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 6 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 5 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 15 & 9 & 15 \\ 9 & 5 & 8 \\ 15 & 8 & 8 \end{bmatrix}$$

(22)

	V_1	V_2	V_3	V_4	V_5
V_1	0	0	0	1	2
V_2	0	0	0	1	1
V_3	0	0	0	2	1
V_4	1	1	2	0	0
V_5	2	1	1	0	0

1(a) $V_1 \rightarrow V_3$ (1:2) $\Rightarrow 3$
 $V_1 \rightarrow V_3$ (1:3) $\Rightarrow 15$

$\cong V_3 \rightarrow V_3$ (1:2) $\Rightarrow 5$